

Numerical Modeling of BRBs using SteelMPF

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BRB Dataset

The calibration database comprises 16 BRB component-level specimens spanning wide ranges of core areas, yielding lengths, and cyclic loading protocols. A subset of these specimens, identified as the training set in Table 1, was used for parameter optimization. This subset includes eight pin-ended braces [1] and one weld-ended brace tested within a strongback-braced frame sub-assembly [2]. The validation set corresponds to a more recent experimental campaign to characterize low-cycle fatigue in BRBs [3].

Table 1. BRB dataset used for model training and validation.

Name	Set	f_y (MPa)	A_{sc} (mm ²)	L_y (mm)	L_y/L_T (-)	Reference
PC160	Train	290	2452	4470	0.79	[1,4]
PC250	Train	290	3845	4557	0.81	[1,4]
PC350	Train	290	5381	4656	0.83	[1,4]
PC500	Train	272	8168	4701	0.83	[1,4]
PC750A	Train	290	11516	4679	0.83	[1,4]
PC750B	Train	290	11529	4557	0.81	[1,4]
PC1200A	Train	290	18406	4704	0.83	[1,4]
PC1200B	Train	290	18464	4605	0.82	[1,4]
PC3SB	Train	290	3225	1930	0.55	[2,5]
A5	Validation	304	3652	3777	0.69	[3]
B3	Validation	307	7226	3048	0.56	[3]
B6	Validation	307	7226	3048	0.56	[3]
B7	Validation	307	7226	3048	0.56	[3]
B8	Validation	307	7226	3048	0.56	[3]
C5	Validation	299	11135	2883	0.53	[3]
CB225	Validation	290	1452	3111	0.70	[6]

BRB Model

The hysteretic behavior of the BRBs was modeled using the Giuffrè–Menegotto–Pinto constitutive formulation [7,8] as extended by Filippou et al. [9] to incorporate isotropic strain hardening and by Kolozvari et al. [10] to allow asymmetric strength and hardening behavior in tension and compression. This formulation is implemented in *OpenSees* as *SteelMPF*, with the following syntax:

```
uniaxialMaterial SteelMPF $mattag $fyp $fyn $E0 $bp $bn $R0 $cR1 $cR2 $a1 $a2 $a3 $a4
```

The BRBs were modeled using a corotational truss element combined with stiff elastic end elements representing the connection regions [5]. As shown in Figure 1, the total BRB length, L , is decomposed into three physical regions: the connection regions of length L_j , the transition regions of length L_t , and the yielding steel core of length L_y . Inelastic deformations are assumed to occur only within the yielding

core, characterized by area A_{sc} , whereas the transition regions, of area $A_t > A_{sc}$, are assumed to remain elastic.

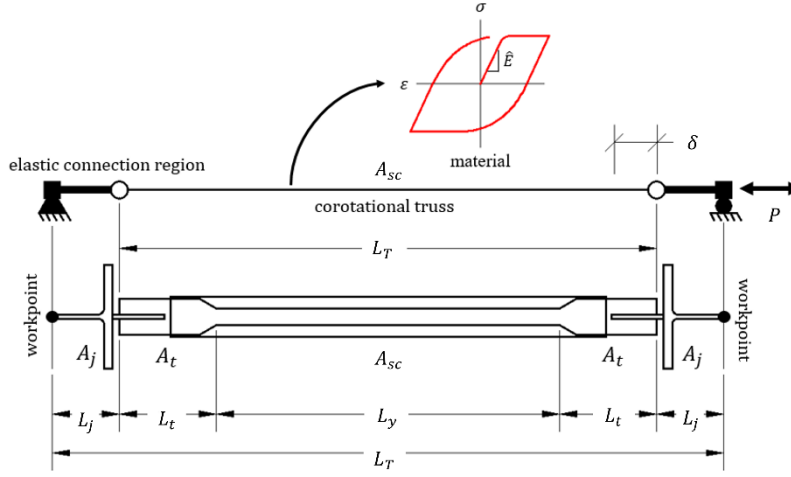


Figure 1. Schematic of BRB model.

To reproduce the global axial stiffness of the BRB without explicitly discretizing the transition regions, an effective stiffness factor, Q , was introduced to account for the series combination of the yielding and transition segments [11]:

$$Q = \frac{1}{\frac{2L_t/L_T}{A_t/A_{sc}} + \frac{L_y}{L_T}}. \quad (1)$$

Using this formulation, the equivalent truss element of length L_T and modified elastic modulus $\hat{E} = QE$ reproduces the combined axial stiffness of the yielding and transition regions. The connection regions are excluded from the computation of Q because they are modeled explicitly.

Results

Table 2 shows the set of parameter values that were found to fit well the response of the train dataset and acceptably generalize the validation dataset. The experimental and numerically simulated force–deformation relationships are shown in Figure 2.

Table 2. Generalized SteelIMPF parameters for the selected BRB dataset.

Parameter	Value	Description
f_{yp}	f_y	Tensile yield strength.
f_{yn}	f_y	Compressive yield strength.
$\hat{E}$	QE	Equivalent elastic modulus, with $E = 200$ GPa and Q from Eq. (1)
b_p	0.0049	Tensile strain hardening ratio.
b_n	0.0233	Compressive strain hardening ratio.
R_0	20.0	Initial curvature parameter.
c_{R1}	0.88	Curvature degradation parameter.
c_{R2}	0.19	Curvature degradation parameter.
a_1	0.037	Compressive isotropic hardening. It shifts the yield envelope by $\Delta\sigma = a_1 f_{yn}$.
a_2	1.0	Compressive isotropic hardening starts after a maximum tensile strain of $a_2 f_{yp} / \hat{E}$.
a_3	0.035	Tensile isotropic hardening. It shifts the yield envelope by $\Delta\sigma = a_3 f_{yp}$.
a_4	1.0	Tensile isotropic hardening starts after a maximum compressive strain of $a_4 f_{yn} / \hat{E}$.

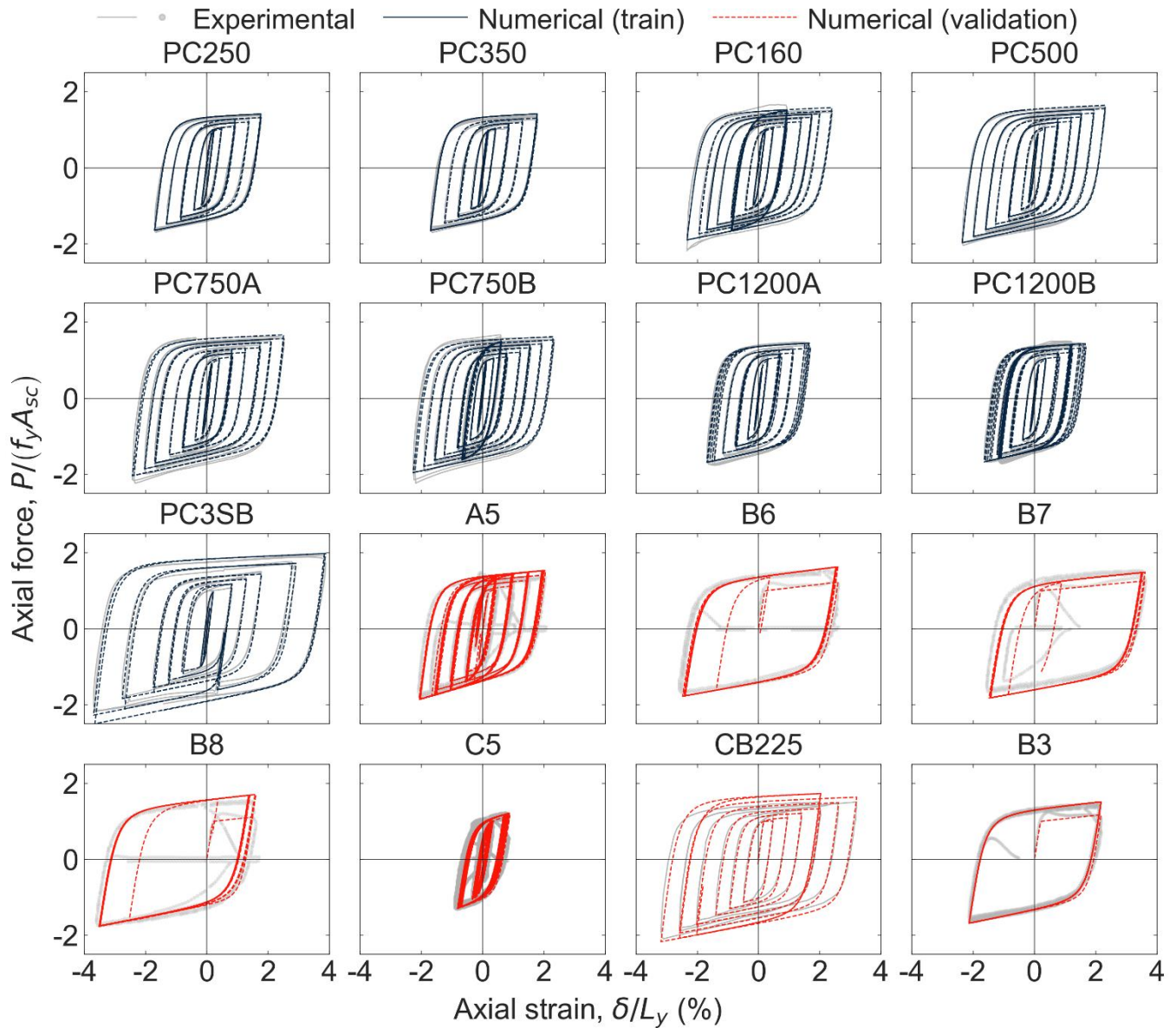


Figure 2. Generalized parameters for modeling BRBs with *SteelMPF*.

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